National Marine Electronics Association

International Marine Electronics Association

Technical Corrigendum TC# 2000 20140710A

ISO Address Claim PGN 060928

This corrigendum updates Fields 3 & 4 of the ISO Address Claim PGN 060928. The note in Field 3 constrained the device instance value range from zero to 252. This corrigendum removes the range constraint of 252, and removes the association between the number of devices with the same class & function code and the configured device instance value (Fields 3 & 4 combined).

The note in Field 3, Device Instance Lower has been replaced by the following:

"The combination of Fields 3 & 4 make up the 8 bit NMEA Instance."

The same note above has been added to Field 4, Device Instance Upper.

Documents Affected

- NMEA 2000 Appendix B, Version 2.000

Changes to NMEA 2000 Appendix B

Replace the following parameter group definition with those accompanying this document:

PGN#	Parameter Group Name
060928	ISO Address Claim

This network management message is used to claim a network address and to respond with device information (NAME) requested by the ISO Request (PGN 059904) or Complex Request Group Function (PGN 126208). This parameter group is always transmitted with a destination address global.

This PGN contains several fields that are Request Parameters. These Request Parameters can be used independently or in any combination.

A node receiving an ISO Request (PGN 059904) for this PGN shall respond by providing this PGN.

If a Complex Request Group Function (PGN 126208) requesting this PGN is received, the receiving node shall respond in the following manner:

- If no Request Parameter fields have been specified with the Complex Request, then the response is to return this PGN, just like responding to the ISO Request (PGN 59904) described above.
- If the Complex Request (PGN 126208) specifies one or more Request Parameter fields, then the response shall be filtered by the one or more fields and field values contained within the request. For example, if the Complex Request for this PGN contained a value for field 2, the Manufacturer's Code, then the device would respond with this PGN, if and only if the device's Manufacturer Code matched the value requested. If the device's Manufacturer code did not match the value requested, then the response would depend on whether the request was global or addressed. A global request containing one or more requested field values that do not match requires no response, while an addressed request containing requested field values, in which one or more do not match, requires a response with the Acknowledge Group PGN (126208), containing the appropriate error codes for each of the requested fields, such as "0x3 = Request or command parameter out-of-range", for the fields that did not match.

ISO Requests for Address Claim parameter group are restricted in address mode and request frequency by the Main NMEA 2000 document, section 8.3.2, "Address to NAME Association Tables".

The ISO Address Claim parameter group is not to be used as a heartbeat message.

Single Frame:	Yes	*Priority Default:*	6	*Default Update Rate:*	milliseconds	*Frequency:*	NA cycles per second
Destination:	Address	*Query Support:*	Required	*Command Support:*	Optional	*ACK Rqmnts:*	Refer to Section 6.4 of ISO 11783-5

Field # *Field Name* Original Reference ID # 63

1 **Unique Number (ISO Identity Number)**
Byte Field Size: *Bit Field Size:* 21
Request Parameter Optional
Command Parameter: Optional

DD173 NMEA Unique Number — Binary number assigned by manufacturer to ensure that the NAME field for each manufactured device is unique, reference NMEA Network Management Section 8.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

2 **Manufacturer Code**
Byte Field Size: *Bit Field Size:* 11
Request Parameter Required
Command Parameter: Optional

DD172 NMEA Manufacturer Code — Assigned by NMEA Committee

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

3 **Device Instance Lower (ISO ECU Instance)**
Byte Field Size: *Bit Field Size:* 3
Request Parameter Required
Command Parameter: Optional

DD201 Generic instance 2 (3-bit) — 0x0 to 0x7 = instance 0 to 7

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

The combination of fields 3 & 4 make up the 8 bit NMEA Instance.

Field	Name			Byte Field Size / Bit Field Size		Request Parameter / Command Parameter
4	**Device Instance Upper (ISO Function Instance)**			*Byte Field Size:*		*Request Parameter* Required
				Bit Field Size: 5		*Command Parameter:* Optional
	DD174 Generic instance (5-bit)			0x00 to 0x1F = Instance 0 to 31;		
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields	

The combination of fields 3 & 4 make up the 8 bit NMEA Instance.

Field	Name			Byte Field Size / Bit Field Size		Request Parameter / Command Parameter
5	**Device Function (ISO Function)**			*Byte Field Size:*		*Request Parameter* Required
				Bit Field Size: 8		*Command Parameter:* Optional
	DD171 NMEA Function Code			Dependent on NMEA Device Class DD170, reference NMEA Class & Function Codes, Appendix B6.		
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields	
6	**NMEA Reserved**			*Byte Field Size:*		*Request Parameter*
				Bit Field Size: 1		*Command Parameter:*
	DD175 Dominant Bit			Set = 0		
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields	
7	**Device Class**			*Byte Field Size:*		*Request Parameter* Required
				Bit Field Size: 7		*Command Parameter:* Optional
	DD170 NMEA Device Class			Dependent on Industry Group DD 168, reference NMEA Class & Function Codes, Appendix B6.		
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields	
8	**System Instance (ISO Device Class Instance)**			*Byte Field Size:*		*Request Parameter* Required
				Bit Field Size: 4		*Command Parameter:* Optional
	DD169 Generic instance (4-bit)			0x0 to 0xF = Instance number 0 to 15;		
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields	
9	**Industry Group**			*Byte Field Size:*		*Request Parameter* Required
				Bit Field Size: 3		*Command Parameter:* Optional
	DD168 Industry Group			0 = Global; 1 = On-Highway; 2 = Agricultural and Forestry; 3 = Construction; 4 = Marine; 5 = Industrial - Process Control - Stationary (Gen-Sets) 6 = Reserved for SAE 7 = Reserved for SAE		
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields	

Marine Industry Group, set = 4

Field	Name			Byte Field Size / Bit Field Size		Request Parameter / Command Parameter
10	**NMEA Reserved (ISO Self Configurable)**			*Byte Field Size:*		*Request Parameter* Optional
				Bit Field Size: resv 1		*Command Parameter:* Optional
	DD001 Reserved field			Variable number of reserved bits, all set to logic "1"		
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields	